



**Uttara
University**

Lab Report

Course Title : Computer Networking

Course Code : CSE0612306

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Cisco Packet Tracer

Cisco Packet Tracer is software that is used to configure network devices and connect them to simulate networking operations. We can configure some virtual networks to this software to simulate connection and data transmission.

There are two types of devices. Network devices and End devices. Network devices are those which connect some end devices to establish a connection between end devices for communication. End devices are those which mainly communicate to each other through a network.

Network Devices

- **Router:** Router mainly connects some networks.
- **Switch:** Switch connects some end devices in the same network to communicate with each other.
- **Firewall:** Protects networks by filtering traffic.

End Devices

- PC
- Laptop
- Mobile
- Server

4 Network Topology

Network topologies are mainly how devices are organized in a network to communicate to each other. There are 4 common topologies used to design a network. Below discussing about those 4 topology:

1. Bus Topology

Bus topology is a type of topology where all the devices are connected to a single cable to communicate to each other. Here the cable is a straight cable that two end points and single are transferred through this cable. Low cost needed to design bus topology.

2. Star Topology

Star topology is a type of topology where all devices are connected to a central device for example to a switch or hub so that all devices can communicate to each other through that central device. Medium cost needed to design star topology.

3. Ring Topology

Ring topology connects all the devices through a circular single cable. This is a little bit same as bus topology but in bus topology that single cable is not in circular shape. There are two end points. But in ring topology the cable is circular. Ring topology is also medium expensive.

4. Mesh Topology

Mesh topology is the most reliable network topology as well as the most expensive to design. Here all the devices are connected to all of each other. So there is a direct connection between every device.